Machine Learning

Deep learning: Regularization for neural networks

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- Therefore, most rules and recommendations are `heuristic` and it requires a good amount of `engineering` to obtain good solutions.

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- The weights and biases are usually initialized as small random values of a normal or uniform distribution.
- Larger starting weights help to avoid losing signal during forward and back-propagation.
- Too large weights, however, result in exploding values during forward propagation.

Regularization

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Parameter penalties

- Similarly to lasso and ridge regression, we introduce an extra term in the objective function that penalizes the size of the parameters:

$$J_{\text{reg}}(\theta) = J(\theta) + \alpha\Omega(\theta),$$

  where $\alpha > 0$ is a tuning parameter and $\Omega(\theta)$ is a norm penalty term, e.g.,

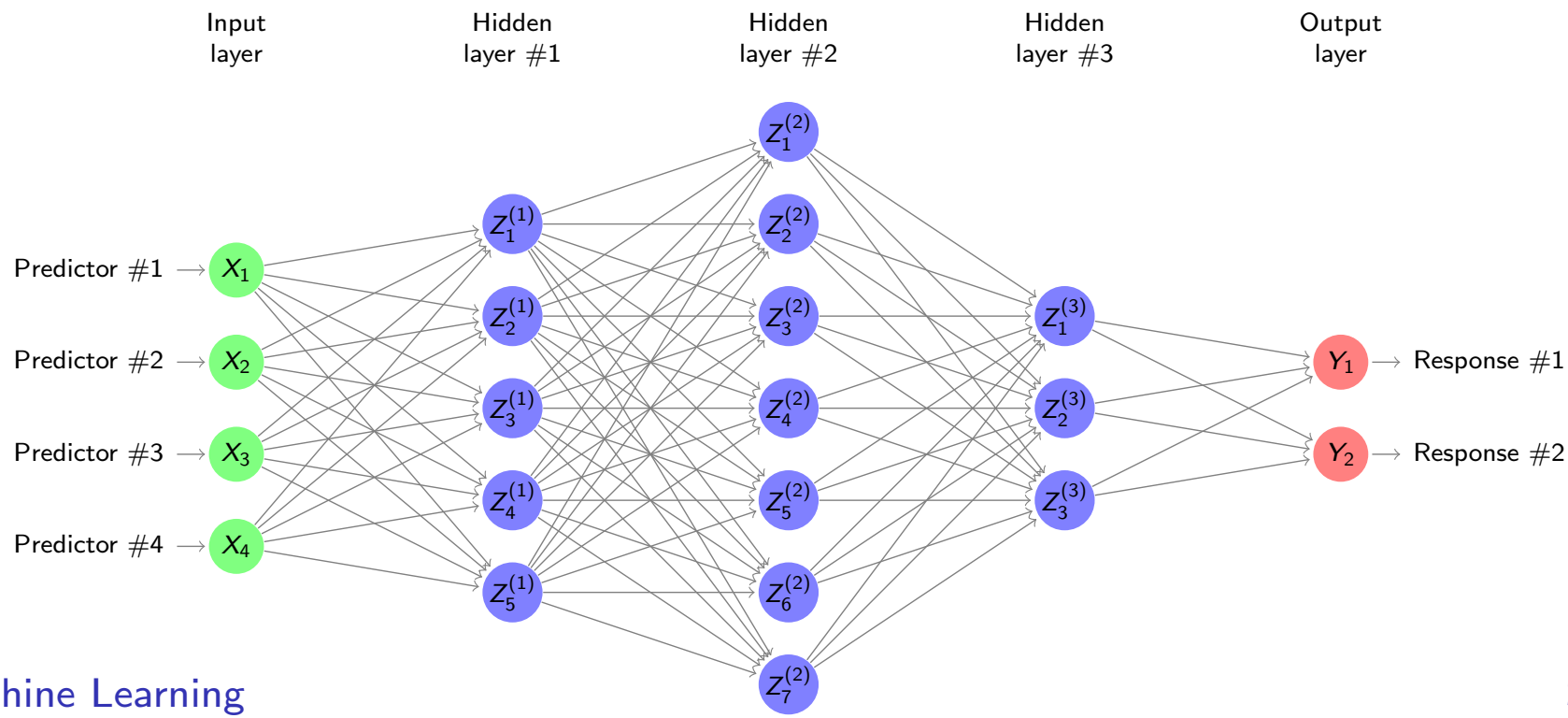
$$\Omega(\theta) = \sum_{i,j,m} {w_{ij}^{(m)}}^2,$$

  or some other norm such as ℓ_1.
- The interpretation is similar as for regularized linear models, and the tuning parameter is also chosen by cross-validation.

Regularization

Dropout

- `Dropout` is a regularization method where in each update step in the gradient descent, the `network structure` is changed.



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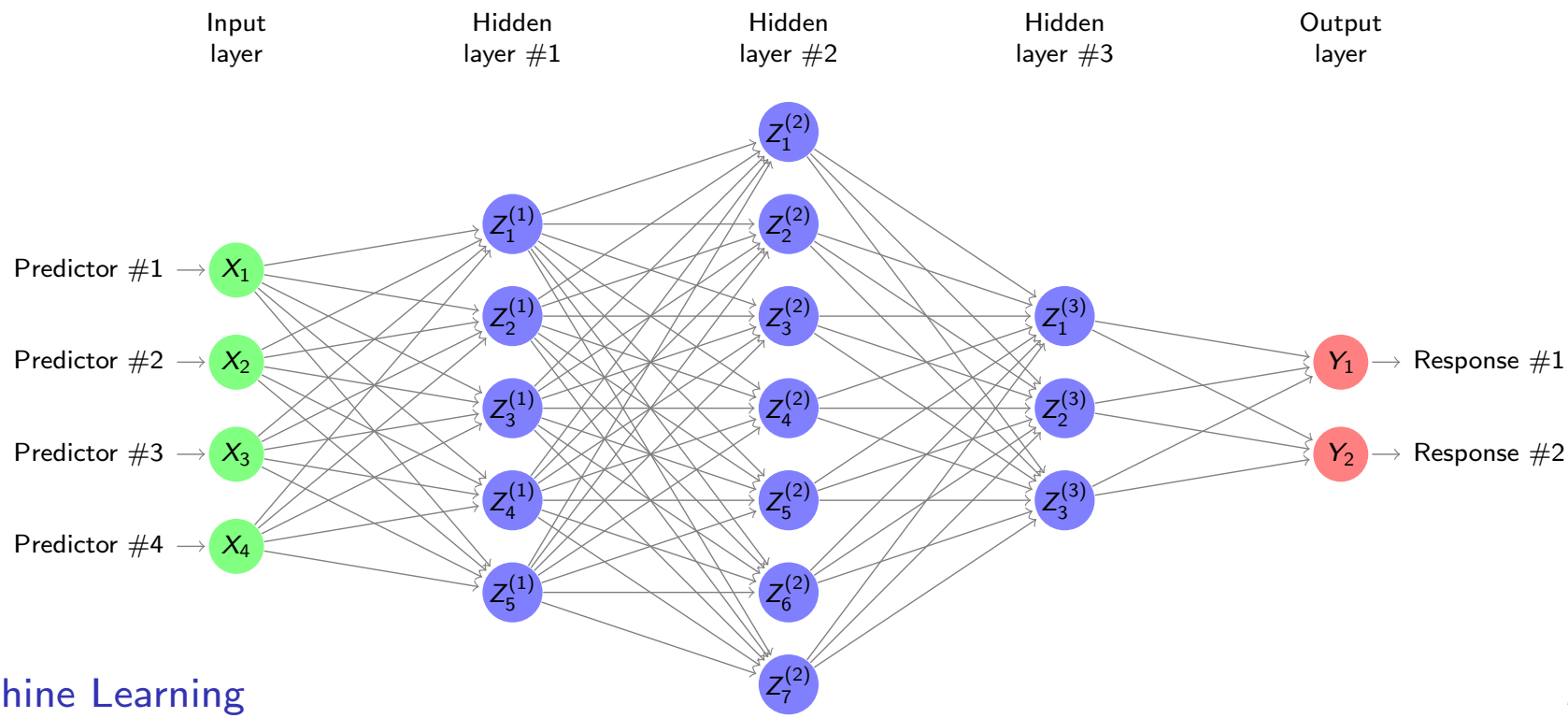
- For `hidden units` this probability is usually $q = 0.5$, and for `input units` $q = 0.2$.



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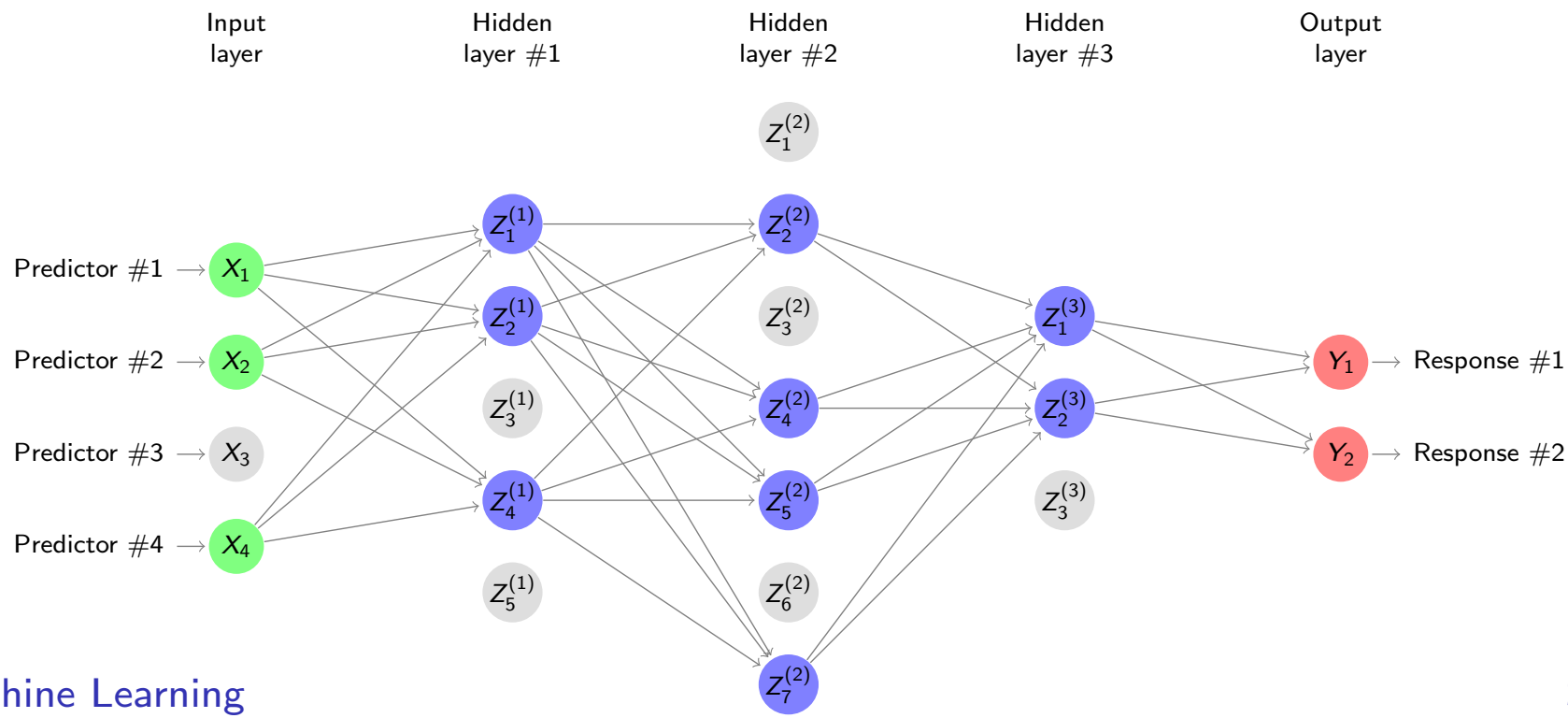

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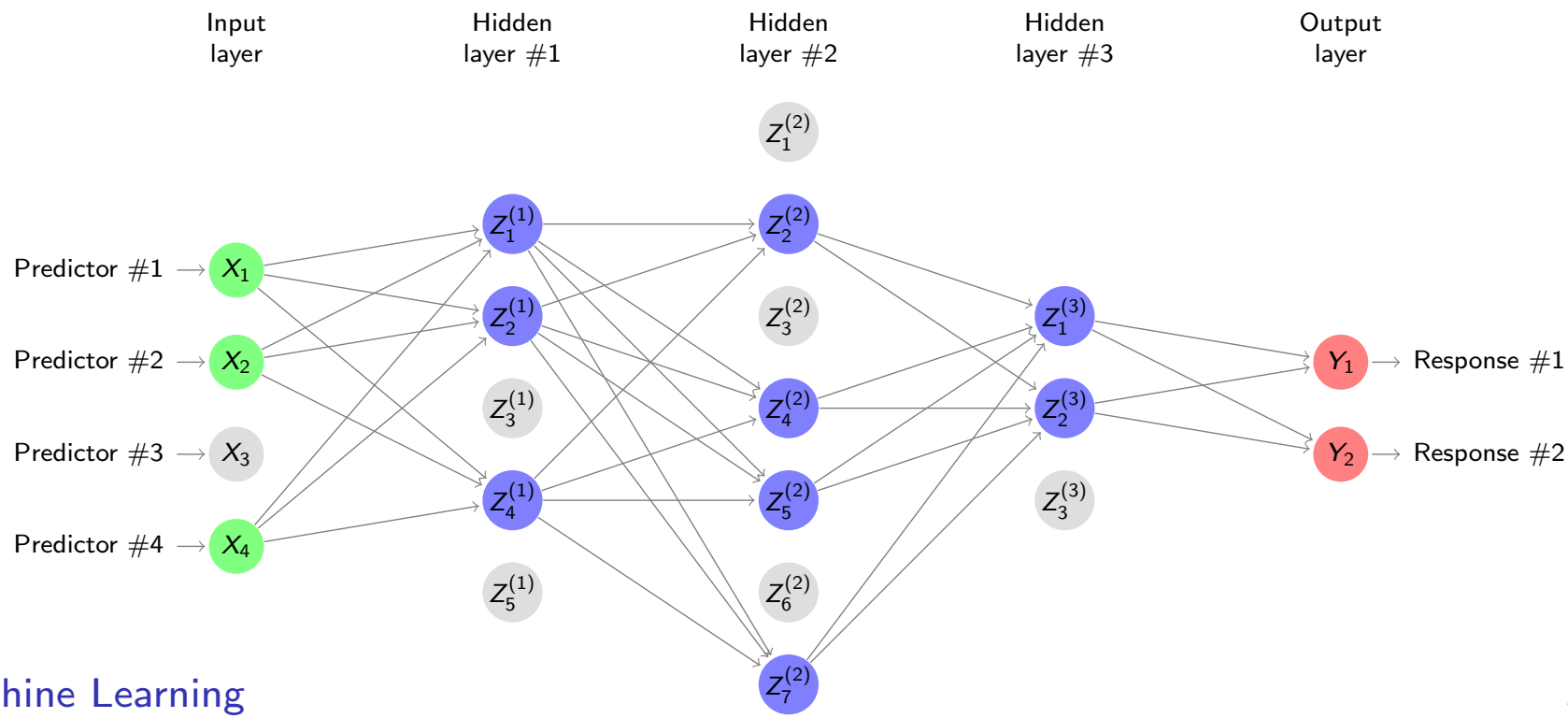

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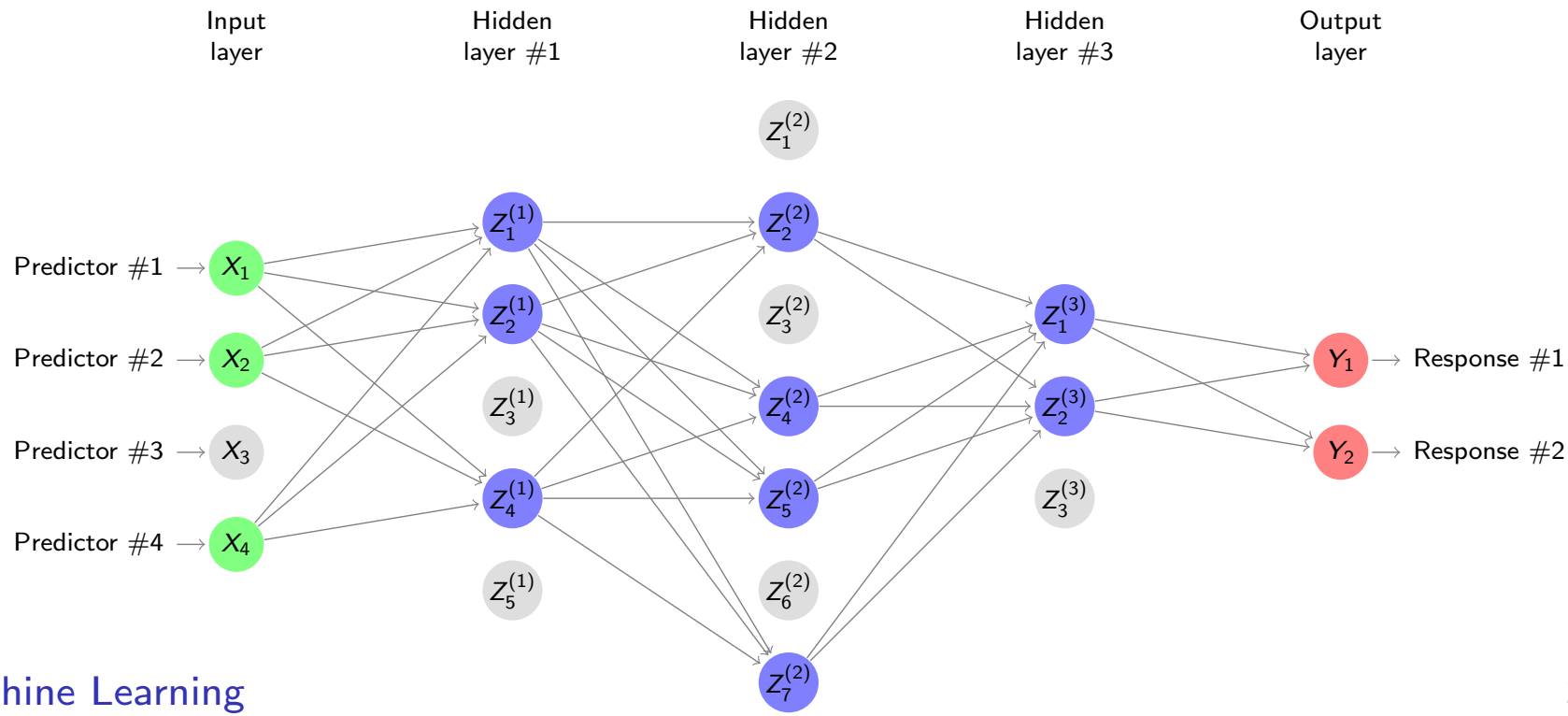
- Dropout is a type of bagging, where the final model is a combination of the fitted weights and the dropout probabilities.



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Regularization for deep neural networks is an active domain of research; the whole Chapter 7 in the book *Deep Learning* by Goodfellow et al. (2016) is devoted to this topic.